

CS361 Assignment 3

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Problem 1

Claim: $\text{Max}\{f(n), g(n)\} = \theta(f(n) + g(n))$

Using θ implies that constants c_1, c_2 and n_0 exist and $c_1, c_2, n_0 > 0$ (strictly positive) AND $f(n)$ exists such that $0 < c_1 g(n) \leq f(n) \leq c_2 g(n)$ for all $n \geq 0$. [Textbook p.94]

Proof:

Since $f(n)$ and $g(n)$ are *Asymptotically Nonnegative* we know that both n_f and n_g exist such that $f(n) \geq 0$ and $g(n) \geq 0$ for all $n \geq \max\{n_f, n_g\}$. We can let $n_0 = \max\{n_f, n_g\}$.

We can assume 2 inequalities trivially, For any $a, b \geq 0$

1) $\max\{a, b\} \leq a + b$

2) $(a + b)/2 \leq \max\{a, b\}$

Assuming $a = f(n)$ and $b = g(n)$ and $n \geq n_0$, we can deduce

$$1/2(f(n) + g(n)) \leq \max\{f(n), g(n)\} \leq f(n) + g(n)$$

Assuming constants $c_1 = 1/2$ and $c_2 = 1$, this matches the definition from θ

$$\text{Max}\{f(n), g(n)\} = \theta(f(n) + g(n))$$

Problem 2

Solve Recurrence Relation $T(n) = 3T(n - 1) + 2$ assuming $T(1) = 1$ and $T(0) = 0$, use unrolling.

$$\begin{aligned} T(n) &= 3T(n - 1) + 2 \\ &= 3(3T(n - 2) + 2) + 2 \\ &= 3^2 * T(n - 2) + 3(2) + 2 \\ &= 3^3 * T(n - 3) + 3^2 * (2) + 3(2) + 2 \\ &= 3^k T(n - k) + 2 \sum_{i=0}^{k-1} 3^i \end{aligned}$$

$$\sum_{i=0}^{k-1} 3^i = 2(3^k - 1/3 - 1) = 3^k - 1$$

$$T(n) = 3^k T(n-k) + (3^k - 1)$$

So using the base case $T(1) = 1$ and $k = n - 1$

$$T(n) = 3^{n-1} T(1) + (3^{n-1} - 1)$$

$$= 3^{n-1}(1) + 3^{n-1} - 1$$

$$= 2(3^{n-1}) - 1$$

$$\text{Thus } T(n) = \Theta(3^n)$$

Problem 3

Guess function for recurrence from Problem 2.

$$T(n) = 3T(n-1) + 2 \text{ assuming } T(1) = 1 \text{ and } T(0) = 0$$

This relation has the form $T(n) = aT(n-1) + b$ with constant b . Solution has

Form $T(n) = A \times a^n + B$ where $a = 3$ and $b = 2$

Substituting into the Recurrence we get

$$A3^n + B = 3(A3^{n-1} + B) + 2 = A3^n + 3B + 2$$

The $A3^n$ terms cancel so and we solve for B, and get $B = -1$

The General Form is $T(n) = A3^n - 1$, and we use $T(1) = 1$. Thus $A = 2/3$

So the Guess function is $T(n) = 2 \times 3^{n-1} - 1$

Proof

Prove for all $n \geq 1$ that $T(n) = 2 \times 3^{n-1} - 1$ assuming recurrence hold for $n \geq 2$

Base: If $n = 1$ then $T(1) = 2 \times 3^0 - 1 = 1$ which is correct.

Induction: If $n \geq 2$ then $3T(n-1) + 2 = 2 \times 3^{n-1} - 3 + 2 = 2 \times 3^{n-1} - 1$

Note: Used outside source for the problem: https://discrete.openmathbooks.org/dmoi2/sec_recurrence.html

Problem 4

A	B	O	o	Ω	ω	Θ
n^k	c^n	YES	YES	NO	NO	YES
$2^{n/2}$	2^n	YES	YES	NO	NO	YES
n^n	$n!$	NO	NO	YES	YES	NO
n^ϵ	$\log^k n$	NO	NO	YES	YES	NO